

Artificial Intelligence Lab – Analytical Problem Solving

Title

Analytical Problem Solving using AI Search Techniques and Intelligent Agents

Objective

The objective of this experiment is to understand how Artificial Intelligence (AI) agents solve real-world and classical problems using problem formulation, search strategies, and intelligent decision-making. The experiment includes the Water Jug problem, Mars Rover agent analysis, 8-Queens problem, OLA Cab problem-solving agent, and Uniform Cost Search (UCS) algorithm.

Question 1: Water Jug Problem

Problem Statement

You are given:

- One **4-gallon jug**
- One **3-gallon jug**

Neither jug has measurement markings. Using only filling, emptying, and pouring operations, obtain exactly **2 gallons** in the 4-gallon jug.

Assumptions

- Fill any jug completely.
- Empty any jug completely.
- Pour water from one jug to another until either:
 - Source jug becomes empty, or
 - Destination jug becomes full.

State Space Representation

Each state is represented as:

(Amount in 4-gallon jug, Amount in 3-gallon jug)

Example:

(0,0)

Initial State

(0,0)

Goal State

(2,x)

where x can be 0–3 gallons.

Solution

Step 4-Gallon 3-Gallon Action

1	0	0	Start
2	0	3	Fill 3-gallon jug
3	3	0	Pour into 4-gallon jug
4	3	3	Fill 3-gallon jug again
5	4	2	Pour into 4-gallon until full
6	0	2	Empty 4-gallon jug
7	2	0	Pour remaining 2 gallons into 4-gallon jug

Final State

(2,0)

Goal Achieved.

Question 2: Mars Rover Agent

i) Percepts

The Mars Rover receives information from:

- Camera images
- Temperature sensors
- Soil sensors
- Rock composition sensors
- Navigation sensors
- Wheel encoder
- Obstacle detector
- Battery level
- Sunlight intensity
- GPS/Position estimator

ii) Operating Environment

The Mars Rover operates in an environment that is:

- Partially Observable
- Dynamic
- Sequential
- Stochastic
- Continuous
- Unknown
- Real-time

iii) Actions

The rover can:

- Move forward
- Move backward
- Turn left
- Turn right
- Stop
- Capture images
- Drill rocks
- Collect soil samples
- Analyze samples
- Send data to Earth
- Recharge using solar panels
- Avoid obstacles

iv) Performance Measure

The rover is evaluated based on:

- Number of samples collected
- Accuracy of analysis
- Distance travelled
- Battery efficiency
- Mission completion
- Communication success

- Obstacle avoidance
- Scientific discoveries

v) Suitable Agent Architecture

Model-Based Goal-Based Learning Agent

Reason

Because it:

- Maintains internal map
- Plans routes
- Learns from environment
- Makes autonomous decisions
- Achieves mission goals

Question 3: 8-Queens Problem

Problem Statement

Place 8 queens on an 8×8 chessboard so that no queen attacks another queen.

Problem Formulation

Initial State

Empty chessboard.

State Space

All possible arrangements of queens.

Actions

Place one queen in a safe position.

Goal Test

No two queens should attack each other.

That means:

- No same row
- No same column
- No same diagonal

Path Cost

Every move has equal cost.

Search Strategy

Backtracking Search

Steps:

1. Place queen in first column.
2. Check safety.
3. If safe continue.
4. Else backtrack.
5. Repeat until all 8 queens placed.

Sample Solution

Q

. . . . Q . .

. Q

. Q . .

. . Q

. Q .

. Q

. . . Q . . .

Question 4: OLA Cab Booking Problem

Suitable Problem Solving Agent

Goal-Based Agent

Reason

The customer's goal is:

Reach destination with preferred cab type.

The agent evaluates different options before selecting the best one.

Pseudocode

Start

Input pickup location

Input destination

Display cab types

Mini

Micro

Sedan

Prime

Shared

User selects cab

Calculate

Distance

Fare

ETA

If cab available

Confirm booking

Else

Suggest another cab

Track driver

Reach destination

Stop

Working

Customer



Enter Pickup



Enter Destination



Select Cab



Check Availability



Calculate Fare

↓

Book Cab

↓

Driver Assigned

↓

Trip Starts

↓

Destination Reached

Question 5: Uniform Cost Search (UCS)

Problem Statement

Find the least-cost path from warehouse **S** to **G**.

Graph

From the given figure

$S \rightarrow A = 1$

$S \rightarrow G = 12$

$A \rightarrow B = 3$

$A \rightarrow C = 1$

$B \rightarrow D = 3$

$C \rightarrow D = 1$

$C \rightarrow G = 2$

$D \rightarrow G = 3$

Initial State

S

Goal State

G

UCS Working

Step 1

Expand

S

Queue

A(1)

G(12)

Step 2

Expand

A(1)

Queue

C(2)

B(4)

G(12)

Step 3

Expand

C(2)

Queue

D(3)

G(4)

B(4)

G(12)

Step 4

Expand

D(3)

Queue

G(4)

B(4)

G(6)

G(12)

Step 5

Expand

G(4)

Goal Found.

Least Cost Path

$S \rightarrow A \rightarrow C \rightarrow G$

Total Cost

$$1 + 1 + 2 = 4$$